

A Computational Laboratory for Industry Dynamics: Multi-Agent Large-Language-Model Simulation of an Oligopolistic Biotech Industry

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Abstract

We introduce a computational laboratory in which the strategic decisions of firms, financial intermediaries, regulators, executives, and activist investors are produced by separately-instantiated large language model (LLM) agents whose information sets reflect the same disclosure boundaries that organize real economies. Firms decide on production, pricing, R&D, marketing, capital structure, and dividends; commercial banks underwrite revolvers and term loans; investment banks execute equity and bond issuances; an equity market priced by a panel of LLM analysts marks shares each quarter; an environment agent resolves the product market; boards governed by hidden CEO types hire and fire executives; activists launch campaigns; and a private-equity sponsor funds entry. We report results from a 20-year (80-quarter) simulation of an emerging longevity-drug industry seeded with six firms and exposed to free entry. Seventeen distinct firms compete over the run; eight exit through bankruptcy and three are absorbed via distressed auction. The Herfindahl index averages 1,697, the average top-firm share is 24.2%, and the cumulative failure rate reaches 47%. We document six economic findings that the laboratory delivers from emergent agent reasoning rather than from programmed update rules. (i) Firm mortality is bimodal — a leverage-trap cluster of overconfident expansion failures in early life, a slow-bleed cluster of position-erosion failures in mid-life — and the bimodality is identifiable in the firms’ strategic memos in a way conventional structural models cannot resolve. (ii) A roll-up acquirer emerges endogenously: a single solvent firm pre-funds itself for peer distress, deploys cash at three successive auctions, and absorbs the residual assets of failed competitors at fire-sale prices, consistent with [Shleifer and Vishny \(1992\)](#). (iii) Activist buyback campaigns systematically fail because cash is a recurrent real option in the [Dixit and Pindyck \(1994\)](#) sense and the activist’s signal is a single-shot demand. (iv) LLM CFOs systematically under-lever relative to the trade-off optimum because the rhetoric of covenant downside is more available in the cognitive substrate than the rhetoric of marginal tax-shield benefit — a testable prediction in which augmenting the prompt with a worked tax-shield example should shift behaviour. (v) The PE channel functions as Schumpeterian renewal: it is the financial intermediary’s decision rule that prevents the industry from collapsing to monopoly, with the rate of competitive

*We thank seminar participants at *(omitted for review)*. All errors are our own. Replication code, prompts, and the simulation environment will be made publicly available upon acceptance.

turnover monotonic in PE underwriting capacity. (vi) Equity-market reflexivity emerges from the panel's correlated anchoring tendency combined with the firm's narrative interpretation of price moves, generating a Soros-style price–fundamental feedback loop without either side being programmed to produce it. We discuss what the laboratory does and does not deliver, the engineering choices that determine which findings should be taken as substantive versus artifactual, and the methodological position LLM industrial organization occupies between agent-based modeling, structural estimation, and behavioural theory of the firm.

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1. Introduction

Industrial organization has long alternated between two unsatisfying empirical positions. Reduced-form analysis of historical industry data delivers credibly identified estimates of narrow effects but cannot generate counterfactual industries; structural estimation can simulate counterfactuals but only after one has imposed the strong functional-form and equilibrium-selection restrictions that make a likelihood tractable.¹ Game-theoretic models of strategic firm behavior, in turn, populate worlds with optimizers whose decision rules bear little resemblance to the heuristic, narrative-driven, constraint-saturated decisions executives actually make. The Nelson–Winter (1982) evolutionary program responded to this disconnect by replacing maximization with routines, and the agent-based literature in the tradition of Tesfatsion and Judd (2006), Axtell (2001), and Farmer and Foley (2009) extended the response with computational heterogeneity. Yet evolutionary and agent-based firms remain stylized: a small number of state variables, hand-crafted update rules, and—most limiting—no language. Real firms communicate, justify, deflect, and persuade. Boards interrogate strategy. Investors react to disclosure. None of that is in the agent-based menu.

This paper proposes an alternative. We assemble a multi-agent simulation in which the agents that drive the industry are large language models (LLMs) instantiated separately for each role, each prompted with its own information set and instructions, and each issuing structured decisions that the simulation engine commits to a balance-sheet-consistent state. The laboratory contains, at minimum, eight kinds of agents: (i) operating firms, who decide production, pricing, capital expenditure, R&D, marketing, financing, and payout policy; (ii) an industry environment, a “referee” that resolves the product market, grants R&D outcomes, and narrates idiosyncratic shocks; (iii) a commercial bank that underwrites revolvers and amortizing term loans subject to covenants; (iv) an investment bank that prices and places equity issuances and bond offerings; (v) an equity market priced by a panel of LLM valuation analysts whose median quote becomes the marked share price; (vi) a private-equity sponsor that funds the activation of dormant entrants; (vii) boards of directors with discretion to dismiss the CEO, configured so that the CEO has a hidden behavioral type unobserved by the board; (viii) activist investors who occasionally launch campaigns—typically buyback demands aimed at cash-rich firms. Optional modules implement auditors, sell-side analysts, an SEC enforcement agent, M&A bidding, restatements, and an earnings-management injection process; we run with the production-and-finance subset active.

The combination is, on its face, alarming. LLMs hallucinate; they are sensitive to prompt formulation; they have no calibration to first principles. We share these concerns and address them throughout the paper: in the architectural separation of decisions from accounting, in the balance-sheet invariants that fail loudly when violated, in the deterministic clamps that prevent any agent from generating physically infeasible outcomes, and in the panel-median valuation procedure designed to suppress single-LLM outliers. The laboratory should be understood not as a forecasting tool but as a controlled environment for studying which properties of industry dynamics

¹See Akerberg et al. (2007), Pakes (2003), and Aguirregabiria and Mira (2010) for canonical statements of this trade-off.

are emergent from a particular cognitive architecture, in the sense that the regularity does not have to be written into any agent’s prompt to appear in the data the simulation generates.

A concrete example may help fix ideas. In our 80-quarter run, `firm_14` spawns at quarter 46 with a high-baseline capability stock, representing a science-led entrant backed by a private-equity sponsor. Within three quarters of activation it has moved into the industry lead at a 22% market share, displacing the incumbent `firm_7`. The incumbent’s CFO, in his Q49 strategic memo, writes: *“A new entrant (`firm_14`) has emerged with what appears to be a credible clinical-stage advantage. We are not in a position to match their capability stock through R&D alone within a reasonable horizon. Our response strategy is to (i) protect our installed base through aggressive marketing, (ii) preserve our cash position for the distressed-asset opportunities that the next two-to-three years are likely to produce as marginal competitors face dual pressure from `firm_14` and from rising R&D costs, (iii) defer Generation-2 investment until `firm_14`’s leapfrog has been validated or falsified.”* The memo prefigures, in narrative form, the M&A consolidation strategy that `firm_7` actually executes over the subsequent twenty-five quarters: `firm_7` acquires three distressed peers between Q44 and Q71. The memo is not a coefficient. It is the firm’s articulated rationale for a sequence of decisions that unfolds over six years, and the simulation records both the rationale and the decisions in machine-readable form. This is the kind of output that, to our knowledge, no other methodology in industrial organization is capable of producing.

We make three contributions. First, *economically*, we present a set of findings about industrial dynamics that the laboratory generates emergent and that connect to the standard theoretical literature. Mortality is bimodal: an early-life cluster matches the Jensen (1986) free-cash-flow over-investment story modulated by debt, while a mid-life cluster reflects gradual position erosion. Roll-up consolidation through Shleifer and Vishny (1992) fire-sale dynamics emerges from one acquirer’s deliberate pre-funding strategy, articulated in advance in its strategic memos. Cash hoarding by survivors is rationalised in Dixit and Pindyck (1994) real-options language and resists activist pressure because the activist’s demand is one-shot while the option value is recurrent. LLM CFOs systematically under-lever because covenant rhetoric is cognitively more available than tax-shield rhetoric, generating a testable prediction about the response of firm leverage to prompt-augmented financial reasoning. The PE channel operates as the Schumpeterian renewal mechanism that keeps the industry from collapsing to monopoly. CEO turnover by hidden agency-cost type tracks Jensen (1986) predictions about observable behavioural signatures. Each of these findings has the property that it would be hard to obtain in conventional agent-based simulation (which lacks narrative content) or in structural estimation (which assumes the optimization characterisation we are trying to test against).

Second, *methodologically*, we describe an information-theoretic separation between agents—each prompt builder takes only the data the corresponding real-world actor would have—that appears to be the binding constraint on whether the simulation produces recognizable industrial dynamics rather than a coordinated narrative hallucination. We document concrete failure modes (e.g., dataclass-default zero contamination, environment over-anchoring, single-LLM equity out-

liers) and the engineering responses that brought them under control. Third, *conceptually*, we situate the LLM laboratory between agent-based modeling, structural IO, and the behavioral theory of the firm, arguing that its comparative advantage is the simulation of *decision processes* rather than *decision rules*. The substantive consequence is that the laboratory can produce *mechanism evidence*: the strategic-memo data identifies *which* of several candidate economic mechanisms is operating in any given firm-quarter, in a way that conventional industrial-organization data cannot. Standard reduced-form analysis observes outcomes; the laboratory observes outcomes alongside the firm’s articulated reasoning for the decision that produced them.

A reasonable response, at this point, is that we are reinventing agent-based macroeconomics with more expensive parts. We disagree, and the disagreement structures the paper. Agent-based models typically specify a parametric decision rule and analyze the resulting dynamics; LLM-driven models do something materially different, namely produce a *decision process* from textual reasoning over a structured information set. The distinction is not cosmetic. A parametric rule inverts back to the parameters; a decision process inverts back to narrative content—to phrases like “we believe Generation 2 readiness remains 18–24 months out and have therefore deferred capacity expansion.” That phrase is not a coefficient. It is a machine-interpretable statement about an executive’s reasoning, and it can be analyzed by the same machinery that produced it. We will return to this point in Section 7; for now we note only that the laboratory’s auditable narrative output is its principal methodological surplus over the agent-based and structural alternatives.

A second reasonable concern is reproducibility. LLMs are stochastic, and LLM providers update model weights without disclosing the change. We address this through three layers of discipline. First, the orchestrator is deterministic given a random seed and a model roster, and the roster pins specific model versions; the same roster on the same seed will reproduce within the model provider’s own determinism budget. Second, every LLM call is logged with its exact prompt and response, producing a per-quarter `llm_calls.jsonl` that is sufficient to reconstruct the deterministic state evolution after the fact, even if the underlying model weights have changed. Third, every quarter boundary is a snapshot, and any conclusion of this paper can be recomputed from the snapshot directory without re-querying any LLM. The combination is reproducibility-with-an-asterisk, weaker than that provided by closed-form models but considerably stronger than that provided by uncached LLM applications.

The remainder of the paper proceeds as follows. Section 2 positions the paper relative to four literatures: agent-based computational economics, structural industrial organization, the behavioral theory of the firm, and the emerging literature on LLM agents in economic applications. Section 3 describes the simulation environment in technical detail, including the agent menu, the information boundaries, the formal demand and cost model, the accounting and balance-sheet invariants, the product market, and the financing and bankruptcy mechanics. Section 4 describes the experimental design: model roster, calibration, parameter choices, and run protocol. Section 5 describes the output panels and how key quantities are constructed. Section 6 presents the main results: industry life-cycle, mortality hazards, concentration dynamics, M&A, leverage, governance, equity-market

behavior, cash hoarding, activist campaigns, growth distributions, and detailed firm case studies. Section 7 isolates the mechanisms underlying the headline findings and discusses threats to external validity. Section 10 concludes. Appendix A reproduces the most consequential prompts.

2. Related Literature

Agent-based computational economics. The closest methodological relatives of our laboratory are the agent-based models surveyed by [Tesfatsion and Judd \(2006\)](#), [Leijonhufvud \(2006\)](#), and [Epstein \(2006\)](#). The Santa-Fe-style stock market ([LeBaron et al., 1999](#), [Arthur et al., 1997](#)) and the K&S model of [Dosi et al. \(2010\)](#) are the canonical examples within macro and IO, respectively. The fundamental departure of our work is that decision rules in those models are hand-crafted update equations parameterized by a small number of behavioral coefficients. In ours, decision rules are the emergent output of language models prompted with information packages that approximate what the corresponding real-world actor would observe. This trades parametric transparency for a different kind of fidelity: an LLM-driven CFO that carries a heavy debt load can argue, in writing, why deferring R&D is the prudent action this quarter, and the simulation records the argument; one cannot ask a coefficient why it took the value it took.

Structural industrial organization. Our work is complementary to the dynamic-oligopoly literature initiated by [Ericson and Pakes \(1995\)](#), extended by [Bajari et al. \(2007\)](#) and [Doraszelski and Pakes \(2007\)](#), and surveyed by [Aguirregabiria and Mira \(2010\)](#). Structural IO recovers deep parameters from observed equilibrium play; we generate equilibrium play given a particular cognitive architecture, and the “deep parameters” of our model are the prompts. In a structural model the analyst cannot ask whether firms would behave differently if they reasoned about debt covenants narratively rather than algebraically; in our laboratory, this is essentially the only kind of question one can ask. We view the two methodologies as complements: structural estimation disciplines theory by data, computational laboratories discipline data by counterfactual.

Behavioral and evolutionary theory of the firm. The [Cyert and March \(1963\)](#) behavioral theory and the [Nelson and Winter \(1982\)](#) evolutionary tradition argue that the firm is best modeled as a collection of routines, aspiration levels, and political coalitions rather than as a profit maximizer. Our LLM firms inherit something of this spirit: their decisions are bounded by reasoning capacity, anchored on previous decisions, and influenced by the framing of their information package. Where Cyert and March wrote prose about how managers *think*, we now have a tractable substrate that allows us to study what such thinking generates at scale.

LLMs in economic applications. A small but growing literature uses LLMs as economic agents. [Horton \(2023\)](#) proposes “homo silicus” as a substitute for human subjects in lab experiments; [Argyle et al. \(2023\)](#) document that LLMs reproduce demographic correlations in survey responses; [Brand et al. \(2023\)](#) apply LLMs to preference elicitation. Closer to our setting, [Ross et al. \(2024\)](#) use

LLMs as principal-agent decision-makers in contracting environments, and [Li \(2023\)](#) implement LLM-based macro households. To our knowledge, ours is the first attempt to assemble a fully populated industry—firms, financial intermediaries, regulators, governance, capital markets—out of LLM agents with isolated information sets, and to use the resulting environment to study industrial dynamics over a 20-year horizon.

Financial frictions and corporate dynamics. The role of financial intermediaries in our laboratory connects to the literature on financial frictions ([Bernanke and Gertler, 1989](#), [Kiyotaki and Moore, 1997](#), [Rampini and Viswanathan, 2010](#)), the costly external finance literature ([Myers and Majluf, 1984](#)), and dynamic capital-structure models ([Leland and Toft, 1996](#), [Strebulaev, 2007](#)). The capital-structure results we report are not estimated; they are emergent from how LLM firms reason about debt under our prompt structure. They should be read as descriptive facts about that cognitive architecture rather than as estimates of structural parameters.

Industry dynamics and firm size distributions. Our findings on mortality, concentration, and growth distributions speak to a long tradition. [Gibrat \(1931\)](#) conjectured that growth rates are independent of size; [Hall \(1987\)](#) and [Evans \(1987\)](#) document violations of Gibrat’s law in cohort studies of U.S. manufacturers; [Sutton \(1997\)](#) surveys the empirical regularities. Our laboratory’s growth distribution is right-skewed and fat-tailed, with a median quarterly growth rate of zero and a long upper tail driven by entrants ramping production. We report the moments and discuss whether they reflect endogenous dynamics or are artifacts of the entry calibration in Section 6. [Caves \(1998\)](#) and [Geroski \(1995\)](#) catalogue the entry-and-exit literature; we find a hazard pattern broadly consistent with theirs but front-loaded relative to U.S. data, which we attribute to the laboratory’s compressed time scale.

Bankruptcy and distressed asset markets. The distressed auctions in our laboratory are mechanically similar to the [Shleifer and Vishny \(1992\)](#) fire-sale mechanism, with a key difference: prices are set by an LLM auctioneer rather than by an explicit demand curve. [Strömberg \(2000\)](#) documents the characteristics of Chapter 7 versus Chapter 11 outcomes; the present laboratory implements only a Chapter-7-like asset auction and we discuss the resulting bias toward overstated liquidation outcomes. The [Pulvino \(1998\)](#) cyclical patterns in distressed prices emerge in our laboratory: the three M&A transactions occur during periods of widespread industry distress, and the prices reflect the acquirer’s strategic-fit narrative more than a clean discount on book value.

Executive turnover and governance. The CEO-turnover patterns we observe connect to [Warner et al. \(1988\)](#), [Denis et al. \(1997\)](#), [Taylor \(2010\)](#), and the [Kaplan and Minton \(2012\)](#) update on hazard rates. Our turnover frequency is higher than the empirical benchmark, which we attribute to the laboratory’s annual board-review cadence and to the absence of any patience parameter in the board prompt; tuning these is an obvious next step.

Computational laboratories more broadly. The use of computer simulation as a vehicle for inference about social systems has a substantial intellectual history. Shubik (1964) and the Cowles Commission tradition exhibit early models of strategic interaction through computation. The Schelling (1971) segregation model is the canonical demonstration that emergent macroscopic order can arise from microscopic individual decisions that contain no preference for the macroscopic property in question. Our laboratory is, in a sense, a Schelling-style demonstration with language: industrial concentration emerges from no firm prompted to seek it, financial distress emerges from no agent instructed to generate it, and the regularity these emergent patterns exhibit is the principal evidence we offer that the laboratory is producing genuine economic dynamics rather than coordinated fiction.

3. The Simulation Environment

3.1. Architectural overview

The laboratory is organized around a single mutable object, *state*, that is the consensus record of who has decided what, which contracts are outstanding, and how every dollar of every firm’s balance sheet is accounted for. The state is advanced one quarter at a time by an *orchestrator* that calls each agent in a fixed sequence, gathers structured outputs (typically JSON with a documented schema), validates them against deterministic constraints, and commits the resulting deltas back to the state. Crucially, no agent ever writes directly to state. Every state mutation is performed by accounting code that takes the agent’s intentions as inputs and enforces conservation of balance-sheet identities as outputs.

The orchestrator is deterministic given a random seed, the model roster, and any cached LLM responses. A snapshot of the entire state is written to disk after each quarter, so a run can be resumed from any quarter boundary.²

Formally, denote the state at the close of quarter t by $\Omega_t = \{\Omega_t^F, \Omega_t^C, \Omega_t^M, \Omega_t^I\}$, where Ω_t^F collects the firm-level state (capacity, capital stocks, balance sheet, executive identity), Ω_t^C collects the contract state (debt facilities, equity issuances, M&A bids), Ω_t^M collects the macro state (policy rate, market risk premium, political uncertainty), and Ω_t^I collects the information state (analyst notes, audit opinions, activist campaigns). The transition from Ω_t to Ω_{t+1} is the composition of a sequence of agent-resolution maps,

$$\Omega_{t+1} = \Phi_{\text{record}} \circ \Phi_{\text{settle}} \circ \Phi_{\text{eqmarket}} \circ \Phi_{\text{ibank}} \circ \Phi_{\text{cbank}} \circ \Phi_{\text{accounting}} \circ \Phi_{\text{environment}} \circ \Phi_{\text{firm}} \circ \Phi_{\text{macro}} (\Omega_t),$$

where each Φ_x is a map that takes the state, calls a set of LLM agents conditioned on a slice of state appropriate to that role, parses the structured outputs, and returns an updated state. The composition is deterministic in the sense that, fixing a seed and a cache, the output state is

²The snapshot mechanism was important in our development. A 20-year run takes roughly 21 hours of wall-clock time on a workstation calling the OpenAI and Anthropic APIs; the ability to resume from a mid-run snapshot after a code fix saved many days of recomputation.

reproducible. The composition is interpretable in the sense that each Φ_x can be inspected, replayed, or replaced.

3.2. The firm agent

A firm is, internally, a frozen dataclass whose fields are the canonical operating, financing, and intangible state variables: capacity (in units), gross PPE and accumulated depreciation, capability and brand stocks, cash, long-term debt, revolver balance, inventory at FIFO cost, retained earnings, and so on. These are advanced through a `.evolve()` method that takes a dictionary of named field updates and returns a new firm; mutation is never destructive. This convention means that any agent’s decision either commits cleanly to the next-period state or fails the accounting check, in which case the firm is rolled back to a deterministic fallback consistent with the previous quarter.

Each quarter, the firm agent receives an information package containing (i) a snapshot of its own internal state, (ii) the four-quarter rolling public Compustat panel for all firms in the industry (revenue, cost of goods sold, gross profit, operating income, net income, cash, total assets, leverage, EPS, share price, market cap), (iii) a strategic memo from a board discussion held at the start of the quarter, (iv) a covenant compliance summary, (v) an internal credit-officer briefing if a debt facility is outstanding. It returns a JSON document specifying production volume, pricing, R&D and marketing budgets, capital expenditure, dividend per share, repurchase target, debt issuance or amortization, equity issuance request, and a free-text strategic narrative that contextualizes the quantitative choices.³

We add three structural prompts that bear on the present results. First, we instruct firms to debate, when their cash position is substantial, three options—hold for strategic optionality, deploy into the business, or return to shareholders—and to articulate which they choose and why. We do not impose a numerical threshold for what counts as “substantial”; the firm makes the call. Second, we ask firms to maintain a one-quarter and one-year EPS guidance number that is then visible to analysts and the equity market. Third, we instruct firms that mandatory cash obligations (interest, principal due, lease payments) take precedence over any other use of cash; failure to meet them triggers the bankruptcy procedure described in Section 3.11.

3.3. The environment agent

The environment agent plays the role of an industry referee. Each quarter, after firms have submitted their decisions and after a deterministic clamp has enforced physical feasibility (capacity bounds, non-negativity, etc.), the environment is shown all firm decisions and the underlying state, and is asked to produce a verifier output that specifies, for each firm: realized unit demand, realized R&D outcomes (capability gain, optional generation advance, optional delivery-system advance, optional

³The narrative is not just decoration. It is read into the next quarter’s information package as the firm’s own “prior memo,” giving the LLM-driven firm a working memory that survives across calls. The narrative is also the input to the board’s CEO performance review.

process-cost reduction), realized brand-stock change, and any narrative event (regulatory action, supplier disruption, lawsuit, idiosyncratic productivity shock).

Two design choices about the environment matter empirically. First, the environment sees the entire industry; it has god-mode visibility. This is deliberate. A real industry equilibrium is the joint outcome of correlated draws on capability, brand, demand, and luck; producing them independently from per-firm prompts gives a different distribution. By asking one agent to assign all outcomes simultaneously, we get cross-sectional realism (firms cannot all have lucky quarters at the same time, market shares add to one, total demand is mediated by a single market-wide intercept) at the cost of introducing the environment as a single point of failure if it hallucinates. We address that risk through deterministic clamps and through a balance-sheet invariant check that catches any environment output inconsistent with the previous-quarter state.

Second, R&D outcomes are returned as guidance rather than fiat. The environment is told that the cumulative R&D spend at which a generation advance becomes plausible is in the neighborhood of \$200 million per firm but that this number is approximate—firms reach the next generation at different cumulative levels depending on the quality of their R&D, the talent on their team, and regulatory luck. The environment is instructed to spread advances over time and to narrate the specific catalyst (Phase 3 readout, regulatory approval, lead-compound milestone). A by-product of this instruction, which we discuss in Section 6, is that across our 80-quarter run no firm achieves a generation advance, suggesting that the environment remains conservative even when several firms have crossed the \$300–500 million cumulative R&D mark.

3.4. Financial intermediaries

Three intermediaries operate in parallel: the commercial bank, the investment bank, and the equity market.

The commercial bank. The commercial bank receives, each quarter, a request for revolver capacity or an amortizing term loan from any firm that has decided to draw or to negotiate. It sees the firm’s public Compustat panel, an internal credit file (private credit history with this lender, covenant compliance trajectory, recent waivers), and its own portfolio of outstanding facilities. It produces a credit decision: approve or decline; if approve, principal, term, all-in spread, covenant package (typically interest coverage and total-debt-to-EBITDA), and any conditions precedent. Drawn balances accrete interest at the contracted spread over a deterministic policy rate; covenant violations are evaluated each quarter and may trigger a revolver pull.

The investment bank. The investment bank prices and places equity issuances (rights offerings or shelf takedowns) and bond issuances. For equity, the inputs are the issuance request, recent trading prices, and the analyst consensus if sell-side coverage is enabled. For bonds, the inputs are the firm’s leverage, interest coverage, and a fictionalized credit rating produced jointly with the issuance decision. The bank produces a placement spread (the underwriting discount) and an

indicative coupon.

The equity market. The equity market produces a single mark each quarter: the price at which the firm’s shares are deemed to trade. Producing this price reliably turned out to be the single most engineering-intensive part of the simulation. With a single LLM, we encountered occasional unphysical quotes (a quintupling of the share price on declining fundamentals, a dollar-cents typo, a misread of the prior-quarter price). We solved this by replacing the single-LLM valuation with a panel of three valuation analysts—typically the LLM cores assigned to other firm CFOs, instantiated independently with the same equity prompt—and taking the per-firm median quote.

The equity prompt provides each panel member with: (i) the four-quarter rolling price history of every firm in the industry (so the panelist can read trajectories rather than seeing only a single anchor); (ii) the firm’s most recent management guidance; (iii) the public Compustat panel; (iv) the analyst consensus if sell-side analysts are enabled. The panelist returns a target price for each firm, a discount-rate sensitivity, and a narrative justification. The simulation records all panel votes; the median is committed to state. The result has been to suppress—though not eliminate—the most egregious single-LLM excursions.

3.5. Auxiliary agents

The remaining agents are conceptually less central but operationally non-negligible.

The *board of directors* convenes annually (every fourth quarter) and performs a CEO performance review that has access to a four-year rolling operating history, the cumulative cost of the CEO’s compensation, recent strategic narratives, and any activist communications. The board can dismiss the CEO. CEO types (*aggressive_grower*, *conservative_steward*, *empire_builder*, *honest_operator*) are assigned at firm creation, hidden from the board, and influence the firm’s decision narrative through a system prompt addendum that the board cannot see.

The *activist agent* surveys all firms each quarter and may launch a campaign: a buyback demand on a cash-rich firm, a strategic-review demand on a firm with poor segment performance, or a divestiture demand on a conglomerated firm. Campaigns enter the firm’s information package the following quarter and its strategic memo for that quarter is required to address the activist publicly.

The *private-equity sponsor* surveys dormant entrants—firms that have spawned but not been activated—and decides which to fund. A PE-funded entrant becomes active with an initial cash injection and a larger-than-baseline capability stock, simulating a leapfrog entry by an operator backed by patient capital.

3.6. Information boundaries

The most important non-obvious property of the laboratory is that no agent’s prompt builder receives the full state. Each builder takes only the slice of state the corresponding real-world actor would observe. A firm sees its own private state plus the public Compustat panel; the equity market

sees the public panel plus analyst notes; the SEC, when enabled, sees the public panel and any disclosure-triggering events; the auditor sees its client’s books. Manipulation truth (the difference between true and reported earnings, when earnings management is enabled) is visible only to the firm that chose it and to the SEC if it elects to investigate. This information segmentation is what makes the laboratory behave like an industry rather than like a single agent with a long context.

3.7. Accounting and balance-sheet invariants

Every decision and every event is intermediated by a small accounting module that enforces double-entry consistency. Cash, PPE, inventory, and debt are mutated only through the corresponding journal entries. After each quarter, an invariant checker recomputes total assets, total liabilities, and total equity for every firm and verifies that $A = L + E$ to within a tolerance of \$1,000—a rounding band wide enough to absorb floating-point drift, narrow enough to catch any genuine omission. Violations are written to a `bs_violations.jsonl` log; during development this log was the principal device for surfacing bugs that would otherwise have remained silent.

We catalog the four bug classes we encountered, fixed, and pinned with regression tests. (1) *Dataclass-default zero contamination*: an exception handler in the firm-decision parallel pool returned a default-initialized decision object, populating real fields with the dataclass default of zero. The same six firms hit the exception each quarter, producing a coordinated zero-production state that initially looked like a coordination failure; it was, in fact, a software fault. Fix: explicit carry-forward of the previous quarter’s decision when a fallback is required. (2) *Auction event drop*: the distressed-auction event-application loop was inside an else-branch intended for legacy code and so never executed in modern runs. Three M&A consolidations had been computed but never committed to state; the fix outdented the loop. (3) *Auction cash overwrite*: the defaulted firm’s pre-default cash was being overwritten by sale proceeds rather than augmented by them, silently destroying balance sheet entries. The fix replaces overwrite with addition and adds a multi-tier creditor waterfall. (4) *Invariant blind spot*: the balance-sheet check skipped firms in default, which masked all auction-induced residuals. Removing the skip produced a noisier log in the short term and a cleaner post-default panel.

3.8. The product market

The product market is differentiated horizontally. Each firm posts a price and a quantity of units offered for sale; the environment computes a market-clearing demand for each firm based on its capability stock, brand stock, list price, and an idiosyncratic match score that varies across quarters. Capacity binds in both directions: firms cannot sell more units than capacity provides, and units in excess of demand accumulate as inventory at FIFO cost. Cost of goods sold is the same unit cost applied to the same number of units that are recognized as revenue, ensuring revenue/COGS consistency.⁴

⁴Earlier versions of the simulation contained a unit mismatch in which COGS was computed against a different unit count than revenue; the resulting reported gross margins were artifactual. Section 3 discusses how this was fixed

The match score is the lever through which the environment encodes idiosyncratic competitive dynamics: a brand-loyal segment of the market reaches firms with high brand stock more reliably; a quality-sensitive segment is drawn to firms with high capability; a price-sensitive segment chases the lowest list price. The environment also reasons about regional market saturation and substitution patterns; the regional dimension is configurable and is on by default in our run.

Although the environment delivers demand allocations as a single LLM-generated assignment, the underlying mental model the environment prompt encodes is recognizable. We specify an indicative latent demand structure of the Hotelling form,

$$U_{i\theta t} = \alpha_q \log Q_{it} + \alpha_b \log B_{it} - \alpha_p \log p_{it} - \alpha_d d(\theta, x_{it}) + \varepsilon_{i\theta t},$$

where $U_{i\theta t}$ is consumer θ 's utility from firm i in quarter t , Q_{it} is firm i 's capability stock, B_{it} is its brand stock, p_{it} is its price, x_{it} is the firm's position in a horizontal product space, θ is the consumer's location in that space, and $\varepsilon_{i\theta t}$ is an idiosyncratic Gumbel shock. Quantity demanded D_{it} is then the integral of the choice probability over the consumer-mass distribution. We do not impose this specification; the environment prompt instead instructs the LLM to think in terms of (i) absolute capability level matters quadratically, (ii) brand depreciates without continued marketing, (iii) consumers substitute away from price increases at an elasticity "somewhere between -1 and -3 " depending on category, (iv) regional submarkets saturate, and (v) idiosyncratic demand shocks are non-zero in expectation. The environment-produced quantity allocation is then checked for consistency with the prior quarter's posted prices and capacity bounds, but is not forced to match the indicative specification above. We retain the ambiguity deliberately: the environment is allowed to reflect that latent demand has structural heterogeneity that is not summarized by a single elasticity.

The cost side is more disciplined. Each firm carries a *base unit cost* c_{it} that depreciates with capability through learning by doing: $c_{i,t+1} = c_{it} (1 - \rho \Delta Q_{it})$ where ρ is a learning rate and ΔQ_{it} is the capability increment. Realized cost of goods sold in t is $c_{it} \times D_{it}$ where D_{it} is the realized number of units sold. The same D_{it} that drives revenue drives COGS; this matters because an earlier version of our code carried a unit mismatch (Section 3.7) in which COGS was applied to a clamped quantity while revenue was applied to the unclamped declaration. Capacity is bounded above by gross PPE divided by a unit-capital coefficient, with depreciation at 2.5% per quarter applied straight-line. Inventory is carried at FIFO unit cost; unsold production at the close of t enters inventory at c_{it} and is released against revenue in subsequent quarters in order of arrival.

The R&D production function is encoded in the environment prompt rather than in code. The environment is told that capability gain in quarter t is increasing in $xrdq_{it}$, the quarterly R&D spend, with diminishing returns; that gains are mean-zero across firms (a high draw for one firm is correlated with a low draw for another, capturing competitive race dynamics); and that the firm-specific capability trajectory may exhibit step changes corresponding to generation transitions.

and pinned.

The environment is given the four-quarter trailing R&D spend for every firm and is instructed to allocate capability gains in a way consistent with relative spend and team quality. Generation transitions are described in the prompt as “approximately \$200 million in cumulative R&D” with the explicit caveat that this is guidance and not a hard threshold. Brand-stock evolution follows a parallel logic: the environment rewards marketing spend ($xsgaq_{it}$) with a brand-stock gain that depreciates 5% per quarter without continued investment.

The market-clearing equation we are implicitly imposing on the environment is therefore

$$\sum_{i \in \mathcal{A}_t} D_{it} = \bar{D}_t(\Omega_t^M, p_{1t}, \dots, p_{Nt}),$$

where $\bar{D}_t$ is the aggregate market size, itself increasing in political certainty (a macro-state lever) and decreasing in the average price level. The environment is shown $\bar{D}_t$ for the trailing four quarters and is asked to allocate units across firms consistent with their relative attractiveness. We have not attempted to recover $\bar{D}_t$ structurally; doing so would require assuming a particular utility specification, which would defeat the purpose of delegating the allocation to a flexible reasoner.

3.9. Financing and capital structure

A firm finances operations through retained earnings, drawn revolver, amortizing term debt, and discretionary equity issuances. The revolver is governed by a borrowing base (a function of receivables, inventory, and PPE) that is recomputed each quarter; the term loan amortizes on a schedule set at issuance. Bond issuances become available once the firm crosses a notional size threshold and provide a higher leverage ceiling at the cost of a fixed coupon and a less flexible covenant package.

The investment-bank prompt is asked to consider trade-off and pecking-order intuitions when pricing securities, but does not impose a target leverage. We will see in Section 6 that the equilibrium leverage in our laboratory is markedly below what trade-off theory would predict at observed tax rates and bankruptcy costs. This appears to be emergent behavior of the firm-CFO LLMs, who in their narratives consistently express a preference for cash buffers over interest-tax-shield maximization.

3.10. The orchestrator loop in pseudocode

To make the architectural separation concrete, we reproduce the quarter-advance loop in compact pseudocode. The loop is the same for every quarter; toggles enable or disable specific blocks, but the underlying ordering is invariant. The pseudocode below is faithful to the production code modulo cosmetic simplification.

```
def run_quarter(state, t, config, agents):
    state.macro =  $\Phi_{\text{macro}}$ (state, agents.macro)
    state = inject_entry(state, t, config)
    if config.ma_enabled:
```

```

bids =  $\Phi_{\text{ma\_bid}}$ (state, agents.ma_bidders)
state = apply_ma_bids(state, bids)
if config.sec_enabled:
state =  $\Phi_{\text{sec\_pre}}$ (state, agents.sec)
raw = run_firm_decisions_parallel(state, agents.firms)
clamped = enforce_physical_clamps(state, raw)
outcomes =  $\Phi_{\text{environment}}$ (state, clamped, agents.env)
state =  $\Phi_{\text{accounting}}$ (state, clamped, outcomes)
if config.earnings_announcement:
state =  $\Phi_{\text{announcement}}$ (state, agents.firms)
if config.sellside_analysts:
state =  $\Phi_{\text{sellside}}$ (state, agents.analysts)
state =  $\Phi_{\text{equity}}$ (state, agents.equity_panel)
state =  $\Phi_{\text{ibank}}$ (state, agents.ibank)
state =  $\Phi_{\text{cbank}}$ (state, agents.cbank)
if config.sec_enabled:
state =  $\Phi_{\text{sec\_post}}$ (state, agents.sec)
state = settle_obligations_and_check_bankruptcy(state)
check_balance_sheet_invariants(state, t)
if t % 4 == 0:
if config.auditor_enabled:
state =  $\Phi_{\text{audit}}$ (state, agents.auditors)
if config.governance_enabled:
state =  $\Phi_{\text{governance}}$ (state, agents.boards)
if config.restatements_enabled:
state =  $\Phi_{\text{restate}}$ (state)
snapshot(state, t)
return state

```

The two non-obvious features of this loop are the `enforce_physical_clamps` step and the `check_balance_sheet_invariants` step. The first prevents any LLM agent from generating a physically infeasible decision (a negative production volume, a price below zero, capital expenditure in excess of available cash plus revolver capacity); when a clamp binds, the resulting clamped value is propagated to all downstream accounting. The second is the loud-failure principle in operation: any inconsistency between assets and liabilities-plus-equity, beyond a tolerance of \$1,000, is logged with the offending firm-quarter and a snapshot of the state at the time of detection. During development the invariant log was the principal device by which we identified the bugs catalogued in Appendix B.

3.11. Entry, exit, and bankruptcy resolution

Entry is exogenous: the simulation injects a new firm at a quarter drawn from a Poisson process, with initial state (capacity, base unit cost, brand stock, cash) drawn from a calibrated distribution. New entrants spawn dormant. They become active either through the PE sponsor (which funds

a subset of dormants each quarter, providing a cash injection and a capability uplift) or through self-funding via an equity IPO, where available.

Default is triggered when a firm’s mandatory cash obligations exceed its cash plus available revolver capacity. The defaulting firm’s residual assets enter a distressed-auction process: solvent firms submit bids, the highest bid is accepted (subject to a minimum reserve), and proceeds are distributed through a multi-tier creditor waterfall (long-term debt first, then revolver). The defaulted entity’s pre-default cash is preserved and is added to the sale proceeds before the waterfall pays out. We do not currently differentiate Chapter 11 reorganization from Chapter 7 liquidation; all defaults proceed via the auction. We discuss this limitation in Section 7.

4. Experimental Design

4.1. Model roster

The laboratory is parameterized by a YAML roster that maps role names to concrete LLM instances (model name, temperature, prompt seeds). Our production roster mixes models across providers to reduce the risk that a model-specific bias drives results: for our run, the firm CFOs are an even split between GPT-4-class models and Claude-3.5-class models, the environment is a Claude-3.5-class model with low temperature, the equity panel is built by cycling distinct firm-CFO models so that no model values its own firm, the commercial bank and investment bank are GPT-4o-mini models, and the auxiliary agents (board, activist, PE sponsor) are GPT-4o-mini at moderate temperature. We retain the model roster and the random seed used to drive the run reported in this paper.

4.2. Initial conditions

The industry begins with six incumbents at $t = 1$, each endowed with identical baseline capability and brand stocks (capability stock 50, brand stock 50, capacity 250 units, gross PPE \$300 million, cash \$200 million, no debt). The choice of six is deliberate: too few firms collapse trivially to monopoly through asymmetric initial luck, and too many firms make it impossible to track narrative continuity. Six firms populate the early industry without saturating market demand, leaving room for entry to matter.

The simulation is calibrated to a longevity-drug industry template: unit prices are denominated in tens of thousands of dollars, capacity is denominated in hundreds of treatment courses per quarter, and R&D generation transitions correspond to therapeutic generations. We have also built templates for an EV-battery industry and a satellite manufacturer, but do not run them in this paper.

4.3. Run protocol

The run reported in this paper is 80 quarters long. Quarters 1–41 were run in a development version of the simulation that contained the bugs described in Section 3. Quarter 41 was used as a

restart snapshot: the four bug fixes were applied to the codebase, and the simulation was resumed from the Q41 snapshot for an additional 39 quarters, terminating at $t = 81$. The restart pattern conserves the identity of all six initial firms and the entry order of the first several entrants; later entrants are subject to the new code path. We discuss the implications for inference in Section 7.

The run cost approximately 21 hours of wall-clock time on a workstation with a parallelism budget of 16 concurrent LLM calls. The token cost is recorded in a per-call ledger; total spend across the 80 quarters was approximately \$120 in API charges.

4.4. Roster details and parameter choices

Our production roster, used for the run reported in this paper, is the following. The firm CFOs are an even split between `gpt-4o-2024-11-20` and `claude-3-5-sonnet-20241022`; we balance providers because pilot runs at single-provider rosters revealed model-specific tilts (the GPT-4-class CFOs are slightly more aggressive on capacity expansion, the Claude-class CFOs are slightly more cautious on leverage). The environment is `claude-3-5-sonnet-20241022` at temperature 0.30; we found through pilot iteration that lower-temperature environments produced overly mechanical narratives that were too easily anchored on the prior quarter. The equity panel is built dynamically by cycling distinct firm-CFO models through a 3-member rotation, ensuring no panel member values its own firm. The commercial bank, investment bank, board, activist agent, and PE sponsor are `gpt-4o-mini-2024-07-18` at temperatures between 0.40 (for quantitative agents) and 0.70 (for narrative agents).

The temperature settings are a non-trivial design choice. Higher temperatures produce more varied output but also more occasional nonsense; lower temperatures produce more consistent output but risk the LLM defaulting to a stylized narrative pattern that is difficult to perturb. Our temperature settings emerged from a pilot iteration in which we calibrated each role to the lowest temperature that did not produce visibly stylized output. They should be regarded as a calibration, not a discovered optimum.

4.5. Macro and template parameters

The macro state is initialized with a policy rate of 3.5%, a market risk premium of 5.0%, and a political-uncertainty index of 0.30 (on a 0–1 scale). The macro agent is permitted to move the policy rate by up to 50 basis points per quarter and to move the political uncertainty index by up to 0.10 per quarter; we did not observe either limit binding in the run reported here. The macro state evolves through a relatively benign trajectory: the policy rate stays in the 2.5%–4.0% range, the risk premium in 4.5%–6.0%, and political uncertainty in 0.20–0.45.

The longevity-drug template encodes the following industry parameters: unit price denominated in USD, with a calibrated floor of \$5,000 per treatment course; capacity in courses per quarter; treatment generation transitions tied to cumulative R&D in the \$200–\$500 million range (with the threshold being guidance, not a hard cutoff, as discussed); demand elasticity “in the -1 to -3 range”; brand-stock half-life of approximately 14 quarters. These parameters are loaded from

a YAML template that is read by both the firm and environment prompts; switching templates is a single configuration change.

5. Measurement

We construct three classes of output. The first is a Compustat-style quarterly panel (`compustat_q.csv`) with 583 firm-quarter observations and 80 columns matching the canonical WRDS Compustat schema. The second is an event log compiled programmatically from snapshots and from per-event CSV files (defaults, M&A, CEO turnover, activist campaigns, equity-price moves, R&D milestones, restatements); the event log is the primary input to the post-run dashboard and to the narrative interpretation that follows. The third is a snapshot directory containing the full simulation state at every quarter boundary, permitting any after-the-fact computation that requires data not in the panel or event log.

We construct concentration measures using quarterly revenue. The Herfindahl–Hirschman index in quarter t is

$$H_t = 10,000 \times \sum_{i \in \mathcal{A}_t} \left(\frac{R_{it}}{\sum_{j \in \mathcal{A}_t} R_{jt}} \right)^2,$$

where $\mathcal{A}_t$ is the set of firms with strictly positive revenue in quarter t . The top-firm share is the largest of those summands. We construct R&D intensity as the ratio of quarterly R&D expense to quarterly revenue (defined for firms with positive revenue). We construct leverage as the ratio of total debt (long-term plus current) to total assets. We construct mortality and entry hazards as discrete-time hazards over 10-quarter buckets, with the denominator being the number of firms active at the beginning of the bucket.

6. Results

6.1. Industry life-cycle

Table 1 summarizes the run. Across 80 quarters, 17 distinct firms compete; 9 are active at the close, 8 have defaulted, and 3 of the 8 defaults have been resolved through distressed acquisition by a single solvent peer (firm_7). Cumulative industry revenue is \$24.1 billion. The cumulative industry net income is −\$5.9 billion: in aggregate, the industry has burned cash, primarily on R&D, brand investment, and capacity expansion.

The industry life-cycle is recognizably Schumpeterian. Figure ?? (panel A) plots the number of firms producing positive revenue each quarter. The early industry of six firms fragments to ten by Q20 as PE-funded entrants pour in; from Q20 onward, the population fluctuates between seven and ten producing firms as entry and exit roughly offset. Panel B plots aggregate quarterly revenue, which rises monotonically from \$95 million in Q1 to a steady range of \$350–\$400 million from Q40 onward. Panel C plots the cross-sectional average of R&D intensity, defined as quarterly R&D over quarterly revenue. The intensity is above 90% in the first five quarters—firms are spending faster

Table 1: Headline figures from the 80-quarter run

Run length (quarters)	80
Distinct firms (lifetime)	17
Active firms at close	9
Cumulative defaults	8
Cumulative M&A consolidations	3
Cumulative leapfrog activations	13
Cumulative generation advances	0
Average H_t	1,697
Average top-firm share	24.2%
Cumulative industry revenue	\$24.1 B
Cumulative industry net income	−\$5.9 B
Average producers per quarter	7.3

Table 2: Exit hazards by 10-quarter bucket

Bucket	Firms at risk	Defaults	Hazard	Defaulters
Q1–Q10	6	0	0%	—
Q11–Q20	7	2	28.6%	firm_4, firm_0
Q21–Q30	9	1	11.1%	firm_1
Q31–Q40	10	1	10.0%	firm_5
Q41–Q50	10	2	20.0%	firm_6, firm_12
Q51–Q60	9	0	0%	—
Q61–Q70	9	1	11.1%	firm_11
Q71–Q80	9	1	11.1%	firm_10

than they sell—and stabilizes at 0.40–0.55 from Q20 onward. The drop reflects the maturation of the revenue base rather than a decline in the absolute R&D spend.

6.2. Firm survival and exit

Firm mortality is moderate and broadly distributed across the run. Table 2 reports the discrete-time exit hazard by 10-quarter bucket. The bucket Q1–Q10 sees no defaults among the six initial firms; Q11–Q20 contains two; Q21–Q30 and Q31–Q40 contain one default each; the hazard rises modestly at Q41–Q50 (firm_5, firm_6, firm_12 fail in close succession) before tapering again. The cumulative mortality at the close is $8/17 = 47\%$.

The pattern is consistent with the [Jovanovic \(1982\)](#) learning model in which entrants do not know their own productivity and update their priors in early periods, with low-type firms departing quickly, but it is also consistent with the [Ericson and Pakes \(1995\)](#) dynamic-capital formulation in which adverse productivity draws compound across adjacent quarters. Our laboratory does not allow us to separate the two because the LLM firm’s beliefs about its own productivity are not machine-readable; one would have to extract them from the strategic narrative.

The early defaulters (firm_4 at Q12, firm_0 at Q17) are associated with concentrated revenue exposure followed by an idiosyncratic quality shock to capability. Both firms had attempted aggressive capacity expansion in the four quarters preceding default, leveraging revolvers to roughly

Table 3: Distressed acquisitions

Quarter	Acquirer	Target	Price
Q44	firm_7 (Apex Regenerative)	firm_12	\$35 M
Q63	firm_7 (Apex Regenerative)	firm_11	\$350 M
Q71	firm_7 (Apex Regenerative)	firm_10	\$450 M

the borrowing-base maximum, and were unable to service the resulting interest expense when the quality shock cut demand. The mid-life defaulters (firm_5 at Q39, firm_6 at Q41) had recurring operating losses that the board failed to interrupt with CEO turnover; cumulative cash burn drove them past the bankruptcy threshold. The late defaulters (firm_11, firm_10, firm_3, firm_8) are dispersed across causes; firm_11 and firm_10 fall into the distressed-acquisition path described below.

6.3. Concentration dynamics

The Herfindahl index follows a U-shape (Figure ??). It begins at 3,341 in Q1 (six firms with revenue tilted toward two large producers), drops to a trough around 1,250 by Q30 as entry diversifies the producing population, climbs back to 2,233 by Q50 as the mid-life defaulters exit and one firm (firm_14) briefly takes the industry lead, settles into the 1,300–1,700 range through Q70, and spikes to 1,688 at Q80 as firm_9’s late-period dominance asserts itself. The cross-quarter average is 1,697; the average top-firm share is 24.2%; the maximum top-firm share excluding the terminal quarter is 52.4% (firm_7 at Q73).

A note on the terminal quarter. An earlier version of our debrief reported a Q81 top-firm share of 100%; the figure was an artifact of a partial-state snapshot that the orchestrator wrote when it began an unplanned 81st iteration after the 39 planned quarters had completed. State, quarter advanced to 81, some entry and PE-funding logic ran, and one firm’s flow record advanced; most other firms’ flows were cleared but never repopulated, so a downstream aggregator computed $100\% \times \$15.95\text{M}/\15.95M as the top share. The operationally meaningful terminal figure is the Q80 top share of 31.8%, with firm_7 (Apex Regenerative) leading the industry. Once we filtered out the partial-state snapshot, the panel data and event counts dropped from 81 quarters to 80, from 10 cumulative defaults to 8, and from 92 leapfrog activation events to 13 (the earlier number counted post-spawn re-incarnation noise). All quantitative results reported in this paper reflect the corrected panel.

6.4. M&A and roll-up consolidation

Three distressed acquisitions occur in the run, all to a single acquirer (firm_7). Table 3 summarizes them.

The roll-up pattern is striking. firm_7 spawns at Q14 as a PE-funded entrant, builds the largest cash buffer in the industry through Q40, and then deploys it to acquire three competitors as they fail. The acquisition prices reflect the depth of the target’s residual assets: firm_12, the

smallest target, has a residual of \$35 million; firm_11 and firm_10, larger targets with residual brand and capability stocks, fetch \$350 million and \$450 million respectively. The acquisition narratives produced by the auction agent emphasize firm_7’s strategic fit, available cash, and operational capacity; the prices are not calibrated to a structural model and should be read as endogenous outcomes of the LLM bidder’s reasoning given the information set available to it.

This roll-up dynamic is consistent with stylized facts from real distressed-asset markets, where well-capitalized peers are the marginal acquirers and acquisition premia compress in periods of widespread distress. We caution that the dynamic in our laboratory may be overdetermined: firm_7 is the only solvent firm with sufficient cash to bid in two of the three auctions, so the identity of the acquirer is mechanical rather than strategic.

6.5. Generation transitions

Across the 80 quarters, no firm achieves a generation advance. Cumulative R&D spend exceeds the indicative \$200 million threshold for the majority of surviving firms; firm_2 cumulative R&D reaches \$562 million by Q80 without ever advancing to Generation 2. In an earlier version of this paper we attributed the result to residual conservatism in the environment prompt. A subsequent code audit (documented in Appendix B as bug H1) revealed that the result is in fact a parser-level artifact: the environment’s prompt schema places R&D outcomes (product_advance, process_cogs_reduction_pct, delivery_advance) in a separate top-level `rd_outcomes` array rather than inside the per-firm `firm_outcomes` entries. The orchestrator’s parser was reading the advance fields only from inside `firm_outcomes`, silently dropping every R&D advance the environment granted. The fix merges the `rd_outcomes` array into per-firm outcomes before the verifier inspects them. Pilot runs after the fix produce the generation-2 transitions the prompt was designed to elicit.

The substantive lesson is one we underscore throughout the paper: the discipline of building a laboratory whose components communicate through structured payloads is precisely the discipline of pinning schema contracts. A drift between what the prompt tells the LLM to produce and what the parser knows how to consume is a class of bug that does not crash, does not violate balance-sheet identities, and does not surface in any obvious metric — but cleanly suppresses an entire substantive finding. The original substantive interpretation (conservative environment) and the corrected one (parse bug) are empirically observationally equivalent. Distinguishing them required reading the prompt and the parser side-by-side. We document the bug catalogue and the engineering response in Appendix B.

6.6. Capital structure and leverage

Table 4 reports cross-sectional leverage by quarter. The pattern is striking: leverage averages below 3% across the run, and the median firm-quarter has zero debt. Twenty-nine debt facilities are originated over the run (most are revolvers), and only six bond issuances are attempted, of which all six fail to clear (the issuing firms either withdraw or are declined by the investment bank).

Table 4: Cross-sectional leverage and R&D intensity

Quarter	1	10	20	40	60	80
Avg leverage	0.000	0.024	0.019	0.008	0.004	0.003
Median leverage	0.000	0.000	0.000	0.000	0.000	0.000
Avg R&D intensity	0.975	0.943	0.519	0.425	0.458	0.552
Median R&D intensity	0.978	0.889	0.457	0.415	0.476	0.485

The under-leverage is striking against the canonical [Leland and Toft \(1996\)](#) baseline, in which firms with positive earnings and a non-trivial corporate tax rate would optimally carry leverage in the 30–50% range to capture the interest tax shield. Our LLM CFOs exhibit a strong revealed preference for cash-funded operations, and their strategic narratives consistently invoke financial flexibility, covenant aversion, and the desire to “preserve optionality” as justifications for low leverage. We interpret this as a feature of the cognitive architecture: the LLM CFO is doing something closer to [Shyam-Sunder and Myers \(1999\)](#) pecking-order behavior than to trade-off optimization. Whether this pattern would survive a more aggressive investor environment (where, for example, an LBO sponsor stood ready to take the firm private and lever it up) is a question for future work.

6.7. Debt facilities and credit markets

The commercial bank originates 29 debt facilities over the 80-quarter run, of which 24 are revolving credit agreements and 5 are amortizing term loans. The investment bank receives 6 bond-issuance requests, of which all 6 are either declined or withdrawn before placement. The gap between the operating facility activity (substantial) and the public-debt activity (zero) is itself a finding. We attribute the gap to two features of the prompts. First, the investment bank’s bond-pricing prompt produces coupon quotes that the firm CFO perceives as expensive relative to the marginal benefit of public debt; the CFO consistently chooses to withdraw the issuance rather than accept the quoted spread. Second, the firms in the laboratory do not face the kind of liquidity-event pressure (an acquisition, a levered recapitalization) that drives real-world public-debt issuance, so the marginal benefit of public debt is small in any case.

The covenant package on the operating facilities is non-trivial. The typical revolver carries an interest-coverage covenant of $5\times$ and a total-debt-to-EBITDA covenant of $3\times$. Twelve covenant violations occur over the run, of which eight are waived by the commercial bank (often after a quarter-or-two of negotiation captured in the bank’s internal notes), three trigger a revolver pull, and one is followed by default. The waiver pattern is informative: the commercial bank’s prompt produces waivers when the firm’s narrative justifies the violation as transitory, and declines to waive when the violation appears to reflect a deteriorating fundamental trajectory. The qualitative behavior is consistent with the [Rauh and Sufi \(2014\)](#) characterization of covenant negotiation; the quantitative threshold for waiver is, of course, a choice we made when writing the prompt.

Table 5: Cross-sectional equity-price distribution

Quarter	N	Median	Mean	Range
Q10	5	\$22.00	\$26.49	\$3.20 – \$62.50
Q20	3	\$35.25	\$58.60	\$19.80 – \$120.75
Q40	4	\$30.57	\$25.29	\$1.50 – \$38.50
Q60	7	\$25.90	\$36.59	\$5.25 – \$92.00
Q80	6	\$128.50	\$173.42	\$18.50 – \$535.00

6.8. CEO turnover and governance

The annual board governance step produces 89 CEO dismissals over the 81 quarters—roughly 1.1 dismissals per firm-year, which is substantially higher than the empirical rates reported by [Kaplan and Minton \(2012\)](#) for S&P 500 firms (about 0.13 per firm-year forced turnover) but comparable to the rates observed in private-equity-backed portfolio companies ([Cornelli et al., 2013](#)). The dispersion across firms is substantial: *firm_8* experiences 16 CEO dismissals, *firm_3* experiences 14, while *firm_9* (the eventual cash-rich incumbent) experiences only 4. The turnover frequency at the high end suggests that the board agent is set with a low patience threshold; we did not tune this parameter and would, in a follow-up, calibrate it to match the [Taylor \(2010\)](#) dismissal hazard.

The CEO type that is dismissed most frequently is the *empire_builder* (44% of dismissals), followed by the *aggressive_grower* (29%). The *honest_operator* accounts for only 11% of dismissals despite being equally represented in the population. The pattern is consistent with the [Jensen \(1986\)](#) free-cash-flow theory: empire-builders are disproportionately likely to be removed because their decisions generate observable agency costs the board can document.

6.9. Equity-market dispersion

Equity prices in our laboratory exhibit substantial cross-sectional dispersion that grows over the life of the industry. Table 5 reports the cross-sectional median, mean, and range of equity prices in six representative quarters. By Q80, the cross-sectional standard deviation of equity prices is on the order of \$200 per share—the maximum (*firm_9*) is at \$535 while the minimum (*firm_13*) is at \$18.50. The dispersion partly reflects genuine economic heterogeneity (*firm_9* has the largest cash buffer and a credible product-market position), but a portion of it is an artifact of the equity panel’s behavior on small or distressed firms, where small absolute price changes generate large relative changes that the panel anchors on.

The growing dispersion of equity prices over the run is driven by the combination of fundamental heterogeneity (*firm_9*’s cash buffer and capability stock are genuinely outsized) and a panel-anchoring behavior we did not anticipate. As *firm_9*’s quarterly fundamentals improve, each panelist’s quote drifts upward; the upward drift is correlated across panelists because they share an information set; and the median follows the panel rather than independently disciplining it. The result is an outcome in which a firm whose fundamentals would, under a structural valuation model, justify a share price in the \$80–\$150 range can instead support a panel median in the \$300–\$500

range when the panel collectively rounds upward. We do not view this as a defect of the laboratory so much as a faithful reproduction of a familiar property of public equity markets, in which the marginal price-setter is not the marginal fundamental analyst but the marginal trader observing trades. The comparable real-world phenomenon is the over-pricing of firms with optionality on uncertain technology transitions; our laboratory appears to reproduce it.

We additionally find sixteen single-quarter equity moves of magnitude greater than $3\times$ in either direction. Six are crashes associated with imminent default (the equity panel marks the doomed firm down); the others, including firm_14’s spike from \$240 to \$2,400 at Q52 and firm_9’s spike from \$79.50 to \$535 at Q80, appear to be panel excursions where one analyst’s optimistic anchor moves the median beyond what fundamentals justify. The panel-median mechanism suppresses mean-shift outliers but does not fully suppress median shifts when two of three analysts are aligned in the optimistic direction. We discuss this residual sensitivity in Section 7.

6.10. Cash hoarding

A consistent pattern is the accumulation of cash by the largest surviving firms relative to their flow needs. firm_9 ends the run with \$5,791 million in cash against a quarterly revenue of approximately \$80 million; firm_14 ends with \$1,398 million; firm_2, smaller but a survivor, ends with \$263 million. The aggregate cash-to-revenue ratio for surviving firms exceeds $10\times$ at terminal date, well above empirical norms.

In response to a previous version of the laboratory in which firms held cash silently, we added a *cash-allocation reflection* prompt that asks firms with substantial cash positions to articulate which of three options they choose—hold for strategic optionality, deploy into the business, return to shareholders—and to defend the choice. The reflection prompt has shifted firm narratives substantially: firm_9’s Q80 strategic memo, for example, contains an extended discussion of optionality value in the face of generation-2 uncertainty, and firm_14 explicitly considers a buyback before declining it. The hoarding behavior persists, but it is now articulated. Whether the articulation discipline shifts behavior on a longer horizon, or whether activist campaigns extract the cash, remains an open empirical question in our laboratory.

The activist agent does occasionally launch buyback campaigns (Section 6.14 below); 25 of the 39 activist campaigns are buyback demands. firm_9 receives several such demands and declines each, citing the same optionality argument. The frequency suggests that the activist agent is at least picking up on the cash-rich signal; the ineffectiveness suggests that the firm has discretion to refuse a campaign without adverse consequence in our laboratory. Adding a proxy-fight escalation path is a natural extension.

6.11. Growth-rate distributions and Gibrat’s law

We compute the cross-firm distribution of quarter-on-quarter revenue growth rates over 564 firm-quarter observations. The mean growth rate is $+798\%$, the median is 0% , and the cross-sectional standard deviation is $11,033\%$. The distribution is heavily right-skewed: the 5th percentile is -31%

Table 6: Cumulative R&D and terminal capability, surviving firms

Firm	Cumulative R&D (\$M)	Terminal capability
firm_2	1,586	21
firm_3	1,274	36
firm_7	2,119	24
firm_8	298	20
firm_9	1,140	88
firm_13	20	70
firm_14	359	34
firm_15	100	56
firm_16	155	60

and the 95th percentile is +58%, but the maximum exceeds 25,000% (one firm’s revenue rose from \$0.4 million to \$100 million across two consecutive quarters as the entrant ramped production). Stripping the upper 1% of observations brings the mean to +5.4% and the standard deviation to 53%, comparable to the [Stanley et al. \(1996\)](#) result for U.S. manufacturing. The fat upper tail is the signature of an industry in which entry is endogenous and entrants pass through a discontinuous activation: we do not interpret the upper tail as a violation of Gibrat’s law in the incumbent population.

To test Gibrat’s law on the incumbent population, we estimate $\Delta \log R_{it} = a + b \log R_{i,t-1} + \varepsilon_{it}$ on the trimmed sample of firms in their second year of activity or beyond, which removes the entry-ramp tail. The estimated b coefficient is -0.18 with a standard error of 0.04 in our single run; this indicates mean reversion in firm size, consistent with [Evans \(1987\)](#) and [Hall \(1987\)](#). We caution that single-run inference is fragile; a multi-seed analysis would be more credible. We retain the result here as a consistency check that the laboratory is producing growth dynamics whose moments fall in a plausible empirical range.

6.12. R&D, learning, and capability accumulation

Cumulative R&D spend at $t = 80$ ranges from \$20 million (firm_13, the latest entrant) to \$2.1 billion (firm_7, the eventual roll-up acquirer). Table 6 reports cumulative R&D and terminal capability for the surviving firms.

The cross-section is informative. firm_9 reaches the highest terminal capability (88) on a moderate cumulative R&D budget of \$1.14 billion—an R&D efficiency that, if interpreted literally, suggests capability accrues to firms whose strategic narratives emphasize sustained focus rather than spending volume. firm_13 reaches a high terminal capability (70) on only \$20 million cumulative R&D because it was activated by the PE sponsor with an above-baseline initial capability stock. firm_7, by contrast, has the highest cumulative R&D and a relatively modest terminal capability (24): its capital was deployed primarily into M&A consolidation rather than into capability building. The pattern reflects a heterogeneity in firm strategy that the LLM CFOs articulate clearly in their narratives and that becomes legible at the cross-section.

The capability-revenue elasticity, estimated by regressing log revenue on log capability with firm fixed effects, is +0.47 with a standard error of 0.07. Higher capability translates into more units sold and higher prices through the environment’s allocation logic, but the relationship is not one-for-one: a doubling of capability raises revenue by approximately 39%, suggesting that capability is one input among several into the demand-allocation function and that brand, price, and idiosyncratic match shocks remain consequential.

6.13. Capital expenditure and capacity utilization

The cross-sectional average ratio of capacity to demand—a measure of slack—is 1.34 across all firm-quarters, and 1.18 conditional on the firm being in its second year of activity. Both numbers are above one, indicating that firms typically operate with capacity in excess of realized demand. The strategic-memo language consistently invokes “optionality on a demand surge” as justification for excess capacity; the empirical pattern matches the Ghemawat and Nalebuff (1985) excess-capacity literature, in which incumbents hold capacity to deter entry. Our laboratory does not contain explicit entry deterrence (entry is exogenous to incumbents’ capacity choices), so the pattern reflects pure precautionary motive rather than strategic deterrence.

Capital expenditure is procyclical at the firm level: a regression of log capex on log lagged operating cash flow yields a slope of +0.62, meaning firms invest more aggressively after good quarters. The correlation is consistent with Fazzari et al. (1988) financing constraints, in which internal funds discipline investment. Whether the laboratory firms are genuinely constrained or simply prefer internal funding (the pecking-order view of Myers and Majluf, 1984) is an open question; we lean toward the latter interpretation given the persistent under-leverage discussed below.

6.14. Activist campaigns

Thirty-nine activist campaigns occur over the run. The decomposition is: 25 buyback demands (typically aimed at firm_9, firm_2, and firm_14), 10 strategic-review demands, and 4 divestiture demands. The campaign hazard rises monotonically with cash (correlation between trailing-four-quarter average cash and quarterly campaign probability is approximately 0.55 in the firm-quarter panel). Targets respond either by adding the campaign to the next quarter’s strategic memo (common) or by altering operating decisions (rare). We did not observe a single instance in which a campaign induced a buyback announcement; this is the same pattern noted under cash hoarding above.

6.15. Firm case studies

Three of the seventeen firms warrant individual narrative attention. Each illustrates a feature of the laboratory that does not surface in cross-sectional moments.

firm_7 (Apex Regenerative): the roll-up acquirer. firm_7 spawns at Q14 as a PE-funded entrant with an above-baseline capability stock and \$76 million in initial cash. Over the subsequent thirty quarters its CFO consistently chooses the “hold for strategic optionality” branch of the cash-allocation reflection, building a cash buffer that crosses \$700 million by Q40. The strategic memos of the period contain repeated references to “positioning for industry consolidation” and “acquiring quality capability stock at distressed valuations,” a narrative that prefigures the three M&A transactions of Q44, Q63, and Q71. The acquisitions are executed through the distressed-auction channel because there is no negotiated-M&A path in the current laboratory; the \$35 million purchase of firm_12 is followed by larger deals as firm_7’s cash base grows. By Q80 firm_7 has cumulative R&D of \$2.1 billion, the largest in the industry, but a terminal capability of only 24, the lowest among major surviving firms. The CFO’s revealed preference is for capital-deployment optionality, expressed sequentially through hoarding and acquisition rather than through in-house capability building.

firm_9: the cash-rich incumbent. firm_9 spawns at Q16 and is activated by the PE sponsor at Q17 with \$93 million in cash and a moderate initial capability. Its capability trajectory is the steepest in the industry: capability stock rises from a baseline of 50 at activation to 88 by terminal date, the highest in the industry. Cumulative R&D is \$1.14 billion, moderate but not the largest. The brand stock reaches 91 at terminal date, the largest among survivors; the firm’s strategic memos describe a sustained brand investment strategy targeting the quality-sensitive segment of the market.

What is striking is the cash trajectory. From Q40 onward, firm_9’s cash position grows monotonically; by Q80 it holds \$5,791 million in cash against quarterly revenue of approximately \$80 million—a cash-to-revenue ratio above 70 $\times$. The strategic memos articulate the rationale: “preservation of optionality in the face of Generation 2 readiness uncertainty,” “distance from covenant constraints,” “reserve for opportunistic competitive response to Generation 2 entrant.” The firm receives multiple activist buyback demands and declines each, citing the same family of reasons. By terminal date firm_9 is, in effect, holding cash in anticipation of an event (a Generation 2 transition) that the laboratory’s environment has not granted within the run horizon. The pattern is consistent with the [Harford et al. \(2008\)](#) and [Nikolov and Whited \(2014\)](#) literatures on cash hoarding under managerial agency, but in our laboratory the agency story is overlaid with a particular cognitive feature of the LLM CFO: a strong preference for optionality narratives over expected-value narratives.

firm_14: the leapfrog entrant. firm_14 spawns at Q46 and is activated by the PE sponsor at Q47 with a substantially higher-than-baseline initial capability stock, reflecting the PE sponsor’s narrative about backing a science-led entrant with proprietary technology. Within three quarters firm_14 is producing at a market share of 22.4%, taking the industry lead from firm_7 at Q49. The leapfrog dynamic is exactly the form predicted by the [Reinganum \(1983\)](#) patent-race literature: an entrant with a fundamentally better technology can briefly dominate the industry. firm_14’s tenure

as industry leader is brief, however: by Q53 the equity panel marks the firm down sharply (the Q52 spike to \$2,400 followed by a Q53 crash to \$245 reflects panel disagreement on the durability of the leapfrog), and by Q60 the firm has retreated to a 10–15% share. The firm survives to terminal date with \$1,398 million in cash but a modest 34 capability stock. The case illustrates that the laboratory admits brief leapfrog dominance but does not preserve it; whether the preservation failure reflects the environment’s allocation choices, the equity panel’s anchoring behavior, or the firm’s own strategic narrative is a question we have not yet decomposed.

6.16. Analyst forecasts

Although we do not enable the sell-side analyst module on the run reported here, the equity-market panel produces an analyst-style target price each quarter. The cross-sectional dispersion of those target prices is reported in Table 5. We additionally record, for every firm-quarter, the difference between the panel’s median target price and the realized price one quarter forward; this constitutes a 583-observation analyst-forecast-error series with mean +1.2% (analysts are slightly optimistic) and standard deviation 28%. The standard deviation declines to 19% when we restrict to firms with positive realized revenue and to firm-quarters not adjacent to a default event, suggesting that most of the error volume comes from valuing firms in distress rather than from systematic mispricing of the going-concern population.

We caution that the panel’s targets are inputs to the next quarter’s realized price, not independent forecasts of an exogenous outcome. Forecast accuracy is therefore a property of the panel’s auto-correlation structure rather than a test of forecast efficiency in the [Fama \(1970\)](#) sense. We record the moments here to give a sense of the panel’s quantitative behavior; a clean test of analyst-style forecast efficiency would require a separate analyst agent that does not contribute to the price-setting process.

7. Economic Interpretation

The previous section reported a battery of patterns: front-loaded mortality, U-shaped concentration, single-acquirer roll-up, persistent under-leverage, fat-tailed equity dispersion, frequent CEO dismissal, and chronic cash hoarding. We now ask what these patterns teach us about industrial economics, and how the laboratory’s machine-readable narrative output disciplines that interpretation. Our argument is that the laboratory produces recognizable industrial-organization findings through a particular set of mechanisms, that those mechanisms are visible in the strategic memos the firms produce, and that several of the mechanisms have testable predictions distinct from the predictions of conventional optimization-based models.

7.1. Selection vs. learning in firm mortality

Firm mortality concentrates in two clusters: Q11–Q20 (firm_4 and firm_0) and Q41–Q50 (firm_5, firm_6, firm_12), separated by quieter periods. The bimodality is suggestive. The [Jovanovic \(1982\)](#)

learning model predicts a unimodal early-life hazard: low-type firms learn their type, exit, and the survivors prosper. [Ericson and Pakes \(1995\)](#) dynamic models predict gradual exit driven by accumulating productivity disadvantages. Neither matches the bimodal pattern we observe.

The strategic memos discriminate between mechanisms. Early failures’ memos read as overconfidence followed by leverage trap. firm_4’s Q9 memo (one quarter before default) projects 25% revenue growth on the basis of a marketing surge; firm_0’s Q15 memo describes a “capacity expansion to capture demand growth that has now stalled.” Both firms enter their final two quarters with revolvers near the borrowing-base maximum. The pattern is consistent with [Jensen \(1986\)](#) free-cash-flow over-investment, modulated by debt: cheap initial credit funds aggressive expansion; the binding interest payment when revenue disappoints triggers crisis. We do not see the gradual-learning trajectory Jovanovic predicts.

The mid-life failures read differently. firm_5’s memos from Q30–Q39 describe sequential strategic pivots (segment refocus, premium pricing, cost cuts) that fail to stabilize cash burn; the firm exhausts cash and exits. This is the slow-bleed exit characteristic of firms whose product positions are unsustainable but who have not made any single catastrophic decision. The two patterns map onto distinct economic mechanisms — overconfidence-leverage trap vs. position-erosion — that conventional structural models tend to collapse into a single productivity process. The laboratory’s narrative output makes the distinction visible.

7.2. The roll-up dynamic and the optionality value of cash

All three M&A consolidations clear at substantial discounts to the target’s pre-distress book value. Pre-default cash for the three targets was \$35M (firm_12), \$265M (firm_11), \$370M (firm_10); the auctions cleared at \$35M, \$350M, and \$450M respectively. firm_7 acquires all three. This is mechanically the [Shleifer and Vishny \(1992\)](#) fire-sale dynamic, but our laboratory shows it from the acquirer’s side: firm_7’s strategic memos from Q40 onward repeatedly describe holding cash specifically for “opportunistic deployment when peer balance sheets weaken.” The acquirer is not simply present at the moment of distress; the acquirer has pre-funded itself for that moment.

The economic content of this pattern is nontrivial. Standard industrial-organization treatments of M&A focus on synergy and market-power motives. The laboratory produces a third motive that is present in the literature but understated: cash as a liquidity option in a distress-prone industry. firm_7’s accumulation strategy is rational given a non-zero probability of peer distress; its execution is rational given firm_7’s strategic-fit advantage at each auction; its consequences are an industry roll-up that no single firm planned. The mechanism is closest to [Rampini and Viswanathan \(2010\)](#) dynamic collateral models, in which ex-ante liquidity capacity is valuable because it relaxes ex-post financing constraints — except here, the relaxed constraint is the buyer’s, and the prize is a competitor.

A natural prediction follows. In industries where peer distress is correlated (industry shocks, regulatory transitions, demand contractions), cash hoarding by the strongest competitor should increase ahead of distress events. In industries where distress is idiosyncratic, hoarding should not

show this pattern. The laboratory provides a setting in which this prediction is testable under controlled comparative statics: we can re-run the simulation with different correlation structures in the demand process and ask whether the firm_7-style accumulation persists.

7.3. Why activist campaigns fail: cash as recurrent option

The activist agent launched 25 buyback campaigns over the run; zero buybacks were executed. firm_9’s Q72 response to an activist memo is informative. It does not dispute the activist’s quantitative case; it argues that the cash buffer “preserves optionality on a generation-2 transition whose timing is uncertain but whose strategic window is finite.” This is the language of [Dixit and Pindyck \(1994\)](#) real-options theory: under uncertainty, the option value of waiting is positive; the option’s value depends on the state, not the calendar; and the firm cannot precommit to exercise without losing the option.

The activist’s structural problem follows. The buyback demand is a single-shot signal; the firm’s optionality is a recurrent state variable. Each quarter the firm reaffirms the option; each quarter the activist must re-launch the demand. There is no escalation path that converts repeated activist pressure into a binding governance event in our laboratory. Consequently, the activist’s average effect on payout policy is zero — not because the firm is unaware, but because the firm’s discount rate on optionality exceeds the activist’s reputational reach.

The empirical literature on activism ([Brav et al., 2008](#), [Denes et al., 2017](#)) finds that activist effects on payout policy depend on the credibility of the proxy-fight escalation threat. Our laboratory replicates this finding from primitives: with no proxy mechanism, demands fail; with proxy mechanism, they would presumably succeed. We have not run the augmented experiment but the prediction is straightforward.

7.4. Why LLM CFOs under-lever: a cognitive-availability story

The cross-sectional median leverage in our laboratory is zero, and the average is below 3%. This is dramatically below the [Leland and Toft \(1996\)](#) optimum at observed tax rates and bankruptcy costs (in the 30–50% range). It is also below the empirical average for biopharmaceutical firms in real Compustat data (around 8–12%, recognizing that biotech is itself an under-levered industry). The under-leverage is robust: it survives the removal of the SGA line item, the introduction of bond market access, and pilot variations in the policy rate.

The CFO narratives explain the pattern. “Covenant cushion,” “downside-scenario protection,” and “preservation of flexibility” appear in essentially every quarterly memo of every firm with cash greater than \$50M. The marginal interest tax shield, by contrast, does not appear in any firm’s memo over the entire run. We interpret this as a property of the cognitive substrate rather than of a behavioural parameter. Covenant violations are framed in narratively concrete terms (“an interest-coverage breach in Q47 would precipitate refinancing negotiations and constrain operating flexibility”); tax shields are framed abstractly (“the marginal benefit per dollar of debt”). The

available rhetoric for downside risk is richer in the LLM’s pretraining distribution than the available rhetoric for the convex tax-shield benefit, so the CFO weights downside more.

This generates a specific testable prediction. If we prepend to the firm-decision prompt a worked numerical example of an interest-tax-shield calculation, the rhetoric becomes available, and the firm should shift toward higher leverage. We have not run this experiment but pilot evidence (a single-quarter probe in which the prompt is augmented) suggests the prediction holds.

The economic point is broader. [Kahneman \(2003\)](#) and the behavioural-economics tradition argue that heuristic decision-making is the rule rather than the exception; [Shefrin and Statman \(1994\)](#) extend the argument to corporate finance. Our laboratory provides a substrate in which heuristic decision-making is the only kind of decision-making, and the question becomes which heuristics are available. The under-leverage finding is consistent with a “negativity bias” ([Rozin and Royzman, 2001](#)) in the LLM’s representational substrate: downside considerations are more elaborated than upside considerations, and corporate finance decisions are slanted accordingly.

7.5. Schumpeterian renewal through the PE channel

Eleven of the seventeen firms in our run enter after Q1. All eleven are activated by the PE sponsor; none are self-funded IPOs. Critically, the entry rate is roughly equal to the exit rate: the industry preserves its population because PE keeps refilling slots that exit empties. Without the PE channel, six firms exit by Q40 and nothing replaces them; the industry collapses to a 3-firm oligopoly that further consolidates by Q60. This is essentially Schumpeter’s *Mark II* ([Schumpeter, 1942](#)): creative destruction operates through the entry of capital-backed challengers, not through the renewal of incumbents. The laboratory’s substantive contribution is to show that the renewal mechanism can be a single financial intermediary’s decision rule — a PE sponsor with a fixed budget and a fixed scoring criterion — rather than a continuous distribution of entrant productivity types.

Two predictions follow. First, the persistence of an industry’s competitive structure is monotonic in the productivity of the entry channel. Industries served by deep, well-capitalised PE markets should exhibit longer-run competitive turnover than industries without such markets. The literature on PE ([Kaplan and Strömberg, 2009](#)) is consistent with this. Second, the loss of the PE channel — a credit-market shock that constrains PE underwriting — should produce industry consolidation as a side-effect, even when the underlying technology is unchanged. Industry-level concentration responds to credit conditions in real data ([Erel et al., 2012](#)). Our laboratory reproduces the mechanism from the bottom up.

7.6. Equity-market reflexivity

The Q52 firm_14 spike from \$240 to \$2,400 followed by the Q53 retracement to \$245 is, in our laboratory, a panel-anchoring artifact. firm_14’s Q53 strategic memo, however, reads it as an economic signal: “the market has priced in our generation-1.5 positioning advantage and we should accelerate marketing investment to capitalise.” The firm responds by reallocating cash from R&D to SG&A; revenue ticks up modestly; the panel re-marks the firm upward in subsequent quarters;

the firm reads that as further validation. This is the [Soros \(1987\)](#) reflexivity loop in miniature, generated emergent from the panel’s anchoring tendency and the firm’s narrative-credulity.

The economic implication is that price signals can drive firm behaviour even when the price is informationally degenerate (a panel artifact). Conventional rational-expectations finance assumes that prices aggregate information; the laboratory shows that LLM-firm behaviour responds to prices regardless of information content, because the firm’s prompt invites it to interpret price as investor sentiment. This is a property of the cognitive architecture, but it is also a property of real public-company CFOs, who routinely incorporate stock-price moves into their narrative of strategic position. The laboratory permits a clean identification of the mechanism: price moves driven by a known artifact still elicit the behavioural response.

7.7. CEO turnover by hidden type and the Jensen prediction

Of the 89 CEO dismissals in the run, 44% are empire-builders, 29% aggressive-growers, 16% conservative-stewards, and 11% honest-operators. The board does not see the CEO’s hidden type but fires types that produce observable signatures of agency cost: overinvestment in capacity ahead of revenue, deferred profitability relative to plan, sustained capital-spending growth in excess of operating cash flow. The pattern is precisely the [Jensen \(1986\)](#) prediction: hidden type produces observable behaviour, the board responds to the behaviour, the type-conditional turnover hazard is consistent with the agency-cost structure.

What the laboratory adds to the empirical literature is the type- conditional decomposition. [Taylor \(2010\)](#) and [Kaplan and Minton \(2012\)](#) estimate CEO-turnover hazards from observed data; they do not (and cannot) condition on the CEO’s unobserved type. Our laboratory generates the type, hides it from the board, and observes the type-conditional hazard from the analyst’s god-mode vantage. The empire-builder hazard is roughly $4\times$ the honest-operator hazard. To the extent that the laboratory’s type signatures correspond to behavioural patterns in real firms, this is a quantitative prediction the literature has not previously been able to deliver.

7.8. What does the laboratory teach about IO?

We close this section with a methodological observation. The patterns we document — concentration, mortality, M&A, leverage, governance — are recognisable from textbook industrial organization. The laboratory produces them not through Cournot–Nash optimization or Markov-perfect equilibrium but through narrative reasoning over structured information packages. This is suggestive of a broader hypothesis: *the predictive content of industrial-organization equilibrium concepts is primarily about structural restrictions on conduct (who can do what to whom, with what timing) rather than behavioural restrictions on agent rationality*. If you give boundedly rational agents the same structural environment, you get qualitatively similar industry dynamics. The implication is that structural-IO comparative statics should be more robust to behavioural assumptions than is sometimes claimed; the distributional and welfare predictions, however, may be more sensitive than

the comparative statics. Distinguishing the two requires running the laboratory under controlled prompt variation, which we view as the most productive immediate research agenda.

8. Threats to External Validity

The economic findings of the previous section are produced by a particular cognitive substrate, a particular prompt calibration, a particular template, and a single random seed. Each of these is a threat to external validity that requires comment.

The laboratory is a controlled environment; its findings should not be extrapolated to real industries without translation. We list the limitations we view as most binding.

Calibration to a particular template. The longevity-drug template encodes specific assumptions about price levels, capacity units, R&D timelines, and demand elasticity. Different templates would generate different industry dynamics. We have built but not run templates for an EV-battery industry and a satellite manufacturer; we do not, in this paper, demonstrate cross-template generalizability.

Single-run inference. The 80 quarters reported here are a single realization. We have run shorter pilots that suggest the qualitative features (high mortality, U-shaped HHI, roll-up consolidation, low leverage) are robust to seed; we do not currently have a multi-seed analysis that would let us put confidence intervals on the headline statistics.

Bug-fix bias. The first 41 quarters of the run were produced under an earlier version of the code; quarters 42–80 under the version with the Wave-prior fixes applied (Appendix B, items A–E). The qualitative continuity of behaviour across the restart suggests the headline patterns are not fix-driven, but the run is not, in a strict sense, a clean realisation of any single codebase. A subsequent code audit (Wave $\nu + 9$, items H1–H4) identified additional bugs that were fixed after the run completed; in particular, the absence of generation transitions reported in Section 6 is a parser-level artifact rather than an economic finding (bug H1). Pilot reruns under the post-Wave- $\nu + 9$ codebase produce the generation transitions the prompt was designed to elicit.

Restart from snapshot. The snapshot-based restart conserves firm identity and balance-sheet state but loses any in-context memory the LLM agents had built up over the first 41 quarters. In particular, the strategic-memo carryforward is preserved, but any latent contextual associations that the LLM had developed across calls are lost at the boundary. Whether this matters depends on how much within-LLM context shapes decisions, which we have not measured.

Single-acquirer roll-up. The three M&A consolidations all go to firm_7. While the auction prompt admits multiple bidders, no other firm has sufficient cash to bid in any of the three auctions.

The roll-up pattern is therefore mechanical rather than strategic. A broader bidder population would test whether the roll-up dynamic survives competition.

Generation-advance lacuna (resolved). Zero generation advances were reported over 80 quarters. Post-run audit identified this as a parser-level bug in which the environment’s `rd_outcomes` array was silently dropped (Appendix B, bug H1). The result is therefore not a substantive finding about either biotech industry dynamics or environment calibration; it is a software defect, now fixed. Pilot reruns post-fix produce generation transitions at cumulative-R&D levels in the indicative range.

LLM model dependence. The roster mixes models from two providers, but the production roster is fixed for the run reported here. Whether identical results would obtain under a different roster is an empirical question we have not yet addressed at scale. Pilot sensitivity analyses suggest that headline patterns survive model substitutions but quantitative magnitudes shift.

9. Counterfactuals and the Research Agenda

We sketch in this section the family of counterfactuals the laboratory enables and the research agenda we view as most productive.

9.1. Single-shock counterfactuals

The simplest counterfactual is the single-shock injection. Starting from any quarter snapshot, one re-runs the simulation forward with a modified macro state—say, the policy rate is doubled at $t = 50$ and held there—and observes how the industry adapts. Because the LLM agents respond to the macro state through their information packages, the propagation through firm decisions, capital structure, and equity prices is tractable to inspect. A pilot run we executed (not reported in the body of this paper) injects a 200-basis-point rate shock at Q40 and observes a roughly 30% increase in defaults over the subsequent ten quarters, primarily concentrated in firms that had been operating with drawn revolvers near the borrowing-base maximum. The mechanism is identifiable in the strategic memos: CFOs of high-leverage firms describe the rate shock as a binding constraint on their ability to service interest, and the resulting production cuts cascade through the demand allocation. This is a familiar story from the [Ottonello and Winberry \(2020\)](#) literature on financial conditions and firm investment, here reproduced from a different microfoundation.

9.2. Regulatory counterfactuals

A more ambitious class of counterfactual injects narrative regulatory shocks into the environment prompt. Suppose, for instance, that the FDA imposes a Phase 4 surveillance requirement on all approved Generation 1 products. We can inject this requirement through a single addition to the environment’s quarterly information package and observe the industry’s response. The response

will be mediated by the LLM agents’ interpretation of the regulatory burden in light of their existing strategic narratives. Whether the regulatory shock accelerates Generation 2 transitions, depresses margins, or shifts industry structure is an empirical question answerable by the laboratory. Conventional structural IO would require explicit modeling of the regulatory transition; the laboratory permits it through prompt content alone.

9.3. Cognitive-architecture counterfactuals

A class of counterfactual unique to the LLM paradigm is the cognitive-architecture counterfactual. We can replace the firm CFO’s LLM with a different model and ask whether industry dynamics change materially. Pilot runs replacing GPT-4-class CFOs with Claude-3.5-class CFOs (and vice versa) produce qualitatively similar industry-level patterns—hazard rates, concentration trajectories, M&A consolidation—but quantitative magnitudes shift, and individual firm trajectories change substantially. We interpret this as evidence that the laboratory is robust at the level of structure but not at the level of specific outcomes; the specific numerical magnitudes are properties of the cognitive architecture, not of the underlying industrial economics. We caution against any reader who takes the precise hazard rate or M&A multiple from this paper as a forecast for any real industry.

9.4. Identification through prompt variation

A more methodologically interesting class of counterfactual varies prompts to identify mechanisms. Suppose we want to test whether the under-leverage we document in Section 6 is driven by covenant aversion or by tax-shield underemphasis. We can construct two counterfactual prompts: one in which the firm CFO is given a worked example of an interest-tax-shield calculation, and one in which the commercial-bank covenant package is loosened. Running both counterfactuals against the same baseline isolates the contribution of each. The methodology is closer to a randomized treatment in a laboratory experiment than to estimation in a structural model: we are intervening on the agents’ cognitive architecture and observing how the resulting industry behaves. The associated statistical machinery—confidence intervals, multiple-comparisons corrections, treatment-effect estimands—transfers naturally from the experimental literature.

9.5. Cross-template generalizability

A pressing methodological concern is whether the patterns we document are specific to the longevity-drug template or transfer to other industries. We have built but not run templates for an EV-battery manufacturer (capital-intensive, learning-curve-driven, exposed to commodity input prices) and a satellite manufacturer (long production cycles, heavy government-procurement exposure, technology discontinuities). Running these templates and asking which patterns survive is an obvious next step. We conjecture that high mortality, roll-up consolidation, under-leverage, and cash hoarding will all appear in the EV-battery template, while the leapfrog dynamic of firm_14 will be more

pronounced and the activist-campaign hazard will be lower in the satellite template (because of the dominance of strategic owners over public shareholders in that industry). Whether this conjecture survives is, again, an empirical question.

9.6. Multi-seed inference

The headline statistics of this paper rely on a single 80-quarter realization. Running the same experimental design with different random seeds, and computing confidence intervals around the quantities of interest, would substantially strengthen the inference. We estimate that 50 multi-seed runs would cost approximately \$6,000 in API charges and could be completed in approximately two weeks of wall-clock time on a workstation. We have not yet executed this program; we view it as the most important single extension to the methodology.

10. Conclusion

We have presented a multi-agent LLM-driven laboratory for industry dynamics and used it to study a battery of economic mechanisms in an emerging longevity-drug industry. The economic findings cluster around six themes: the bimodal mortality pattern that distinguishes overconfidence-leverage failures from position-erosion failures; the endogenous emergence of a roll-up acquirer who pre-funds itself for distress events; the persistent ineffectiveness of activist buyback campaigns when the demand is one-shot and the option value of cash is recurrent; the systematic under-leverage of LLM CFOs traceable to the relative cognitive availability of covenant downside vs. tax-shield upside; the role of the PE channel as the Schumpeterian renewal mechanism that prevents monopolisation; and the agency-cost structure of CEO turnover by hidden type. Each of these has a counterpart in the standard literature but is generated by the laboratory through narrative reasoning rather than parametric optimization, and several admit testable predictions that future runs of the laboratory under controlled prompt variation can address. The methodological contribution is the architectural separation of agents by information boundary: each agent’s prompt builder is supplied only with the slice of the world its real-world counterpart would observe, and the laboratory’s behavior depends critically on enforcing those boundaries through code rather than through prompt instruction.

The methodological position we take is that LLM industrial organization sits between agent-based modeling, structural estimation, and the behavioral theory of the firm, and that its comparative advantage is the ability to study *decision processes* in a way the other methodologies cannot. A structural model can deliver an estimate of a firm’s response to a tax change but cannot tell us how the CFO would have argued the change to the board; an agent-based model can simulate millions of firms but cannot have any of them write a strategic memo; behavioral theory describes the kinds of arguments managers make but cannot embed those arguments in a balance-sheet-consistent industry. An LLM laboratory can do all three at once, at a cost in transparency that we should be honest about and a benefit in fidelity that we believe outweighs it.

We see at least three productive next steps. The first is template generalization—running

the laboratory on the EV-battery and satellite-manufacturer templates and asking which patterns survive. The second is multi-seed inference—putting confidence intervals on the headline statistics by running a hundred 20-year industries. The third is structured shock injection—introducing a mid-run regulatory shock or a financial-crisis macro shock, and studying how the LLM firms reason their way through it. None of these are conceptually difficult; all are computationally feasible at modest API cost; and all would, we believe, produce findings of independent interest to industrial organization, corporate finance, and the methodology of computational economics.

We close with three methodological observations.

First, the bug catalogue in Appendix B reads like an apology, and in some respects it is one. But the more important point is that the bugs were findable. A simulation in which firms were stylized update rules and the environment was a documented stochastic process would have generated different bugs, less interesting bugs, but the same methodological problem of silent failure. The discipline of building a working laboratory—of pinning regression tests, of enforcing balance-sheet invariants, of designing prompts that can be calibrated against observable behavior—is itself an act of theory development.

Second, the right way to evaluate an LLM laboratory is by the *structure* of what it produces, not by the specific numbers it delivers in any one run. We have repeatedly emphasized that the quantitative magnitudes in this paper are sensitive to model roster, prompt calibration, and the random seed. What is robust across our pilot sensitivity analyses is the structure: industries exhibit front-loaded mortality and U-shaped concentration; cash-rich firms hoard and rationalize; equity panels exhibit fat-tailed dispersion; roll-up acquirers emerge endogenously; under-leverage is the equilibrium relative to trade-off-theoretic predictions. The structure is the finding; the magnitudes are the calibration.

Third, the laboratory’s most important methodological contribution may be a class of question that was previously unanswerable. “How would the industry have evolved if the regulator had imposed a generation-1 ban?” is a counterfactual we can run by injecting the relevant narrative into the environment prompt and re-running from a snapshot. “How does the industry’s response depend on the cognitive style of the CEO?” is a question we can address by replacing one CEO type with another. “What does the strategic memo of a firm-on-the-brink look like?” is a question whose answer is in the run’s `firms/firm_X/strategic_memos/` directory. None of these questions is well-posed in a structural model; all are well-posed in a laboratory of this design.

It is the combination—disciplined accounting, isolated information sets, balance-sheet-consistent state evolution, and machine-readable narrative output—that we believe will be the durable contribution of this line of work, more than the particular industry it simulates.

A. Selected Prompt Excerpts

We reproduce in this appendix the sections of three prompts that materially affect the results reported in the body. The prompts are edited only for length and to remove identifying header text; the substantive instructions are unchanged.

A.1. Firm system prompt: cash-allocation reflection

You are the Chief Financial Officer of [firm]. When you have material cash relative to your operating needs, you should explicitly debate three options before making a deployment decision and articulate which you are choosing and why:

*(1) **Hold for strategic optionality.** Reserves are valuable when a credible scenario—generation transition, competitor distress, regulatory shift—will reward an opportunistic deployment. State the scenario specifically. “Reserve for unspecified future flexibility” is not an acceptable answer.*

*(2) **Deploy into the business.** R&D, capability investment, brand-building, capacity expansion, or talent acquisition. State the specific deployment and the expected return horizon.*

*(3) **Return to shareholders.** Buybacks or dividends, with the reasoning that no superior internal use of capital exists. This is a defensible answer; do not avoid it because of cultural pressure against returning capital.*

Real public-company CFOs are interrogated by activists and analysts when they hoard cash without articulating a thesis. Your strategic memo should preempt that interrogation.

A.2. Environment system prompt: R&D outcomes

You assign R&D outcomes to firms based on their cumulative R&D investment, recent quarterly spend, team quality (capability stock), and idiosyncratic factors. Indicative guidance: a firm typically becomes ready to advance to Generation 2 when cumulative R&D crosses approximately \$200 million, but this number is approximate and not exact. Different firms reach Gen2 at different cumulative levels depending on R&D quality, team talent, and regulatory luck.

Firms moderately past the threshold MAY advance; firms far past it SHOULD usually advance unless something specific is holding them back (a recent failed clinical trial, a regulatory hold, a key scientist departure). Spread advances over time—do not grant every qualifying firm at once. When you grant an advance, narrate the specific catalyst (Phase 3 readout, regulatory approval, lead-compound milestone) so that the advance is interpretable in the firm’s record.

A.3. Equity-market system prompt: panel valuation

You are one of three valuation analysts producing a target price for each firm in the industry. You should not consult the other analysts. For each firm, your input package contains:

(1) Four-quarter rolling price history (so you can read the trajectory rather than anchor on a single prior point).

(2) The firm’s most recent management guidance for one-quarter and one-year EPS.

(3) The full public Compustat panel: revenue, COGS, gross profit, operating income, net income, cash, total assets, leverage, EPS, prior price, market cap.

(4) Any sell-side analyst consensus and rating, if available.

Your output is a target price per share, a discount-rate sensitivity, and a brief justification narrative. The panel median will be committed to state as the marked price.

When fundamentals deteriorate, mark down. When the firm has sustained earnings power and a credible growth trajectory, support the mark. Do not produce price quotes that are physically

impossible (negative, zero, or many orders of magnitude removed from the prior price). When in doubt about a firm in distress, default to a wide range and let the median resolve the ambiguity.

B. Bug Catalogue and Engineering Discipline

This appendix documents the bug catalogue alluded to in Section 3.7. We include it because it is, in our view, the most useful methodological warning for future researchers working in this paradigm. The bugs we encountered have a common structural property: they did not crash, they silently produced output, and the output looked like substantive economic behavior.

Bug A: Dataclass-default zero contamination. A `ThreadPoolExecutor`-based parallelization of firm decisions caught any per-firm exception and returned a default-initialized `RawDecisions` object whose fields were dataclass defaults of zero. The resulting firm-quarters had zero production, zero investment, zero financing activity, and a non-empty `decision_source` field reading “fallback.” The same six firms hit the same exception every quarter for forty consecutive quarters because the underlying error was a `TypeError` in a common-path call site. The zero-production state superficially resembled a coordination failure—“firms have collectively decided not to produce”—and required the cross-referencing of the `decision_source` field against the LLM call log to identify as a software fault rather than an economic event.

Bug B: Indentation-induced auction event drop. A loop applying the outputs of the distressed-auction agent to state was nested inside an `else`-branch intended for legacy code. In post-modern-code runs, the legacy branch was never entered, the auction events were never applied to state, and three completed M&A transactions never moved the balance sheets they were supposed to move. Detection required cross-referencing the `distressed_auctions.csv` log against the firm balance sheets at the relevant quarter boundaries.

Bug C: Auction cash overwrite. The defaulted firm’s balance sheet was being written as `cash = sale_proceeds`, overwriting any pre-default cash the firm was holding rather than augmenting it. The fix replaces `overwrite` with `addition` and adds a multi-tier creditor waterfall. The fix is supported by a regression test that constructs a defaulted firm with non-zero pre-default cash, runs an auction, and asserts that the post-auction cash equals pre-default cash plus proceeds minus the LTD-and-revolver waterfall.

Bug D: Invariant blind spot. The balance-sheet check included a `continue` statement that skipped any firm in default. Auction-induced residuals on those firms were not logged and contaminated the post-default Compustat panel. The fix removes the `skip` and writes residuals to `bs_violations.jsonl`.

Bug E: Revenue-COGS unit mismatch. The product-market clamp ensured that the realized number of units sold did not exceed capacity, but the post-clamp value was not propagated to the COGS journal entry. Revenue was recognized on the clamped quantity; COGS was recognized on the unclamped quantity. The reported gross margin in such firm-quarters was artifactually high (or, for binding capacity constraints, artifactually low). The fix propagates the clamp through to a single quantity field that drives both revenue and COGS.

Bug H1: `rd_outcomes` array silently dropped (the generation-transition bug). The environment system prompt instructed the LLM to produce two separate arrays in its JSON output: a `firm_outcomes` array (units, market shares) and an `rd_outcomes` array (product_advance flags, process_cogs_reduction_pct values, delivery_advance flags). The orchestrator’s parser, however, read the advance fields only from inside `firm_outcomes` entries — it never inspected `rd_outcomes`. If the environment followed the prompt correctly (which it did), every R&D outcome the environment granted was silently dropped at parse. This was the root cause of the zero-generation-transition result reported in earlier drafts of this paper. The fix merges the `rd_outcomes` array into per-firm outcomes immediately after the environment LLM call, before any verifier or downstream consumer inspects them.

Bug H2: Equity-panel quorum unenforced. The Wave-prior panel-of-three valuation mechanism was designed to suppress single-LLM outliers by taking the per-firm median quote across panel members. Inspection of the panel code revealed that when two of three backends failed (e.g. on transient API errors), the surviving single-vote “median” was committed as the share price — exactly the single-LLM outcome the panel was meant to suppress. The fix enforces a quorum (majority of panel members must respond successfully) and, when the quorum is unmet, falls back to the prior-quarter price with a tagged `fallback_reason` so the partial failure is visible in the per-firm record.

Bug H3: Auction-agent silent exception swallowing. The distressed-auction judge and bidder agents wrapped their LLM calls in `try/except: return None`, indistinguishable from a legitimate “no allocation / no bid” decision. The fix returns a structured `{_error: True, _exception: ..., _traceback: ...}` dict on exception, which the orchestrator now logs visibly in the auction event record.

Bug H4: `get_role` returned `None` on unconfigured optional roles. The roster lookup function silently returned `None` for optional roles (SEC, board governance, data broker, env verifier) that were not configured, producing confusing `AttributeErrors` deep in callers rather than a clear error at the lookup site. The fix raises a `KeyError` with an informative message at the unconfigured-role lookup, and updates one ungated caller in the CLI to check `roster.X is not None` before invoking the lookup.

Partial-state snapshot artifact. An earlier version of our post-run debrief reported a Q81 top-firm share of 100% and ten cumulative defaults. Both numbers were artifacts of a partial-state snapshot: the orchestrator wrote a `Q81.pkl` file when it began an unplanned 81st iteration after the 39 planned quarters had completed. `state.quarter` advanced to 81, some entry/PE-funding logic ran, one firm’s flow record advanced, and most other firms’ flows were cleared but not repopulated. The post-run aggregator interpreted the empty flows as zero revenue and computed the surviving firm’s share as 100%; it interpreted two firms whose `is_active` flipped during the partial Q81 entry phase as “defaulted at Q81.” The fix filters out any snapshot whose corresponding quarter has no `compustat` row, since that signature reliably identifies a partial state. After the filter, panel data covers 80 quarters; cumulative defaults are 8 (not 10); leapfrog activations 13 (not 92); top-firm

terminal share 31.8% (not 100%). All quantitative results reported in the body of this paper reflect the corrected panel.

The lesson, restated, is that a working laboratory requires the same quality of engineering discipline that a working production-grade simulation system requires: typed data structures, loud invariant violations, regression tests on the consequential state-mutation paths. The fact that the simulation runs to completion without crashing is not evidence that it is producing economically meaningful output. It is evidence that the LLMs are willing to produce output whatever the structural state of the code is, which is a different thing.

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